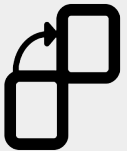
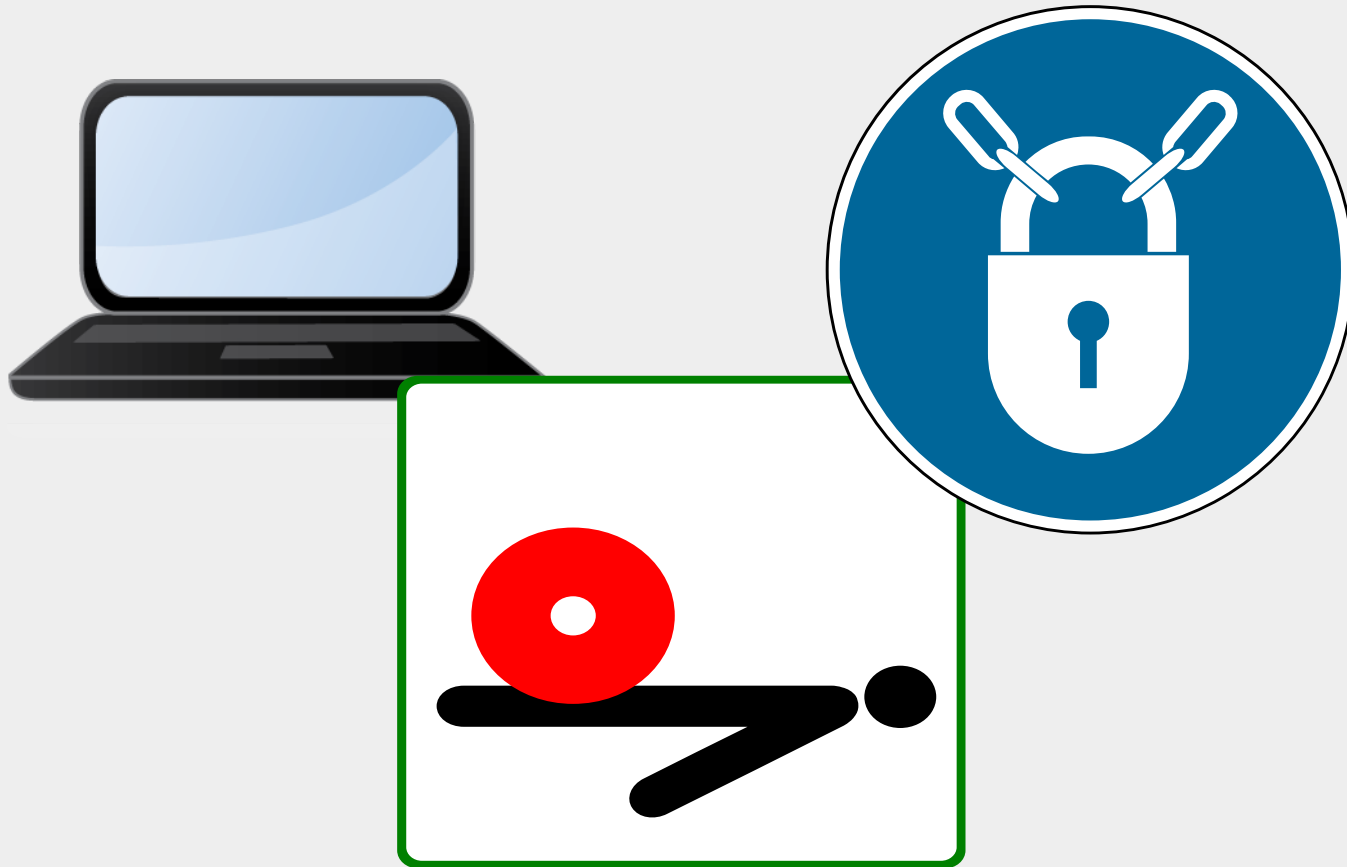


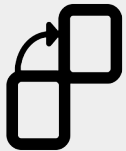
Cryptography

For Hackers and Engineers



What I Do

- Software Engineer at Microsoft (Pittsburgh)
 - Day to Day, I write code for a cloud storage product
- Tech educator at chaintuts
 - I create free and open content about cybersecurity, cryptography, digital currencies, computer science topics – videos, articles, code, book
- Volunteer at the Cryptocurrency Certification Consortium (C4)
 - I help educate people about digital currencies and security
- Outside of work – usually find me doing something active or learning something new



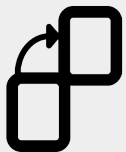
Why This Talk? (And Disclaimers)

- Cryptography is neat! It underpins a lot of what we do as engineers, security professionals, etc.
- It's actually easy to get wrong...catastrophically
- This talk is all about explaining the most common broad categories and use cases...so you can understand where and why we use crypto
- Disclaimer: I am not a cryptographer...just a guy that loves tech that uses it
- Implementing something important? Use a library...and consult a professional



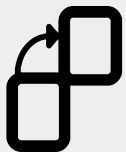
Broad Categories of Note

- Hash functions – Your blender for data
- Asymmetric Encryption – I don't know you...let's share secrets anyway
- Symmetric Encryption – Your key is my key is our key



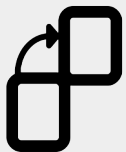
Hash Functions

- One-way, deterministic function that takes any data as input, gives a fixed-length output
- One-way – cannot go backwards from output to input
- Deterministic – same input gives the same output every time
- Fixed-length – for example, 256 bits (SHA-256)



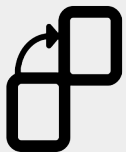
Cryptographic Hash Functions

- Have an important property – *preimage resistance*
- In practical terms, this means there's nothing in the hash output that gives any idea as to what the input is
- Doing so is computationally infeasible
- In order to find out what the input was, you just have to guess until you get the right output



Hash Function Use Cases

- Password Hashing (and cracking)
 - Database doesn't store raw password, stores the hash
 - When you log in, you provide plaintext password. Server hashes and checks that the hash output matches the database.
 - This is why password cracking exists – perhaps you have a database of hashes and need to find the original passwords for account takeover



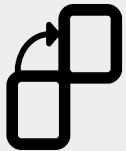
Hash Function Use Cases

- Blockchains
 - Transaction/block IDs – it's a block “chain” because the new block references the past block hash
 - Proof-of-work mining – Miners have to hash tx data and random values until they get a value that is less than the “difficulty target” – hash interpreted as 256-bit number
- File integrity verification – “fingerprints”
 - Used by you to prove a file download has remained intact in transit



Asymmetric Cryptography

- System that uses a private key (kept private) and a public key (shared) to encrypt data, or for verifiable digital signatures
- Key feature is the ability to share keys (public keys) and establish secure communications without a prearranged key share
- This also means we can use the public-private key relationship to sign messages, and allow public verification of that signature



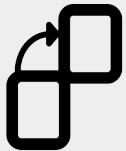
Asymmetric Crypto Use Cases

- TLS/SSL (HTTPS) – Talking between client and server securely
 - Can't have a secret flash drive exchange for every website we visit...
 - Instead, use asymmetric crypto to exchange information and arrive at a shared secret (symmetric) key (or exchange a secret key, but that isn't common anymore)
 - TLS is complicated – but at a surface level – we're using public keys to encrypt information that only one party can decrypt, so we can exchange information in the open



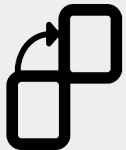
Asymmetric Crypto Use Cases

- Passwordless Authentication
 - Instead of a password hash, server stores a public key
 - When you log in, you sign a message using your private key – the server can verify this, proving you are who you say you are
 - Any leaks of the pubkey would be harmless (in the sense that it cannot be cracked the way a password hash can)



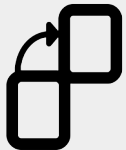
Asymmetric Crypto Use Cases

- Cryptocurrency Transactions
 - Every address is basically a public key (hash)
 - You send coins to that address in a transaction
 - When the owner wishes to spend, they sign a transaction using their private key and post that signature to the blockchain – others can validate the signature with the pubkey, without revealing the privkey
 - This creates a “chain” of signatures showing coin ownership



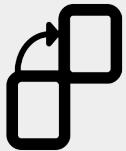
Symmetric Cryptography

- System that uses a single key for encryption and decryption – either random or a “stretched” password
- Advantages
 - Speed! Much faster than asymmetric algorithms
 - Relatively small key size, modern is typically 128-256 bits
- Useful for encrypting large amounts of data, embedding in chips, etc.



Symmetric Crypto Use Cases

- Full disk encryption – Bitlocker, Veracrypt, etc.
 - Allows fast encryption and decryption of data – so fast you don't notice it
 - Typically provide a passphrase on startup, or perhaps key is stored in TPM on device
- Single file encryption – 7zip, LibreOffice encrypted file
 - Uses a stretched password as key
- Used in TLS to do the actual encryption



Q&A - Ask Me Anything!

